

Table 1. Six-method comparison on five selected curve cases

Historical 32-internal-knot experiment | Errors without square roots

Curve source	Method	K	MSE (x 10 ⁻⁵)	MaxSE (x 10 ⁻⁴)	Time (ms)	Pass
Synthetic	Ours	19	3.694	3.674	1191.40	Yes
	Park	32	6.451	4.621	103.96	No
	Liang	32	5.119	2.050	92.61	No
	Dung	25	4.802	3.156	1683.14	Yes
	Kang	26	4.882	2.112	1292.60	Yes
	Luo	30	4.871	2.789	1294.84	Yes
UJI handwriting	Ours	3	4.767	1.766	829.14	Yes
	Park	5	4.183	4.150	18.51	Yes
	Liang	5	4.721	2.766	27.39	Yes
	Dung	4	3.097	1.510	189.76	Yes
	Kang	3	4.297	4.696	3450.89	Yes
	Luo	3	4.035	1.792	8508.51	Yes
Natural Earth	Ours	13	4.385	3.807	1710.59	Yes
	Park	12	2.608	2.451	69.56	Yes
	Liang	13	4.236	4.359	99.82	Yes
	Dung	14	2.192	3.076	1097.47	Yes
	Kang	13	4.018	4.036	2005.55	Yes
	Luo	13	3.729	4.266	8258.45	Yes
USGS contours	Ours	13	4.196	2.188	710.59	Yes
	Park	18	3.596	3.634	143.21	Yes
	Liang	16	4.330	2.895	41.62	Yes
	Dung	14	4.870	4.607	700.94	Yes
	Kang	14	4.075	4.152	1815.06	Yes
	Luo	14	4.973	4.106	6488.42	Yes
Industrial offsets	Ours	5	2.843	1.582	540.77	Yes
	Park	6	3.801	2.331	35.58	Yes
	Liang	6	3.820	1.686	14.50	Yes
	Dung	6	2.719	1.093	325.56	Yes
	Kang	8	1.831	2.522	3132.55	Yes
	Luo	6	4.176	1.591	6822.16	Yes

Selected illustrative cases (one per source), not dataset averages. K denotes internal knots; capacity is 32.
MSE and MaxSE are mean and maximum pointwise squared Euclidean errors in normalized coordinates (no square root).
Pass requires MSE <= 5e-5 and MaxSE <= 5e-4. Total time includes measured numerical postprocessing.
Historical protocol: all six methods use insertion repair; only Ours additionally uses post-pruning. Baselines are repository adaptations.
Bold marks the minimum in each numerical column within a case (ties included); lower error alone does not imply the best simplification.